

Son Preference in Parental Time: Do Families Invest More Care in Male Children?

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Abstract

India's strong son preference is well-documented in fertility decisions, sex-selective abortion, and inheritance patterns. Using India's 2024 Time Use Survey (10.2 million observations from 139,489 households), we examine whether son preference extends to differential parental time investment. We construct household child gender composition from individual records and link to parental time use. We find evidence that household child composition influences parental childcare time allocation. The analysis controls for family size, parental age, education, employment, and state fixed effects. These patterns suggest that son preference may operate not just in who is born but in how children are raised, with potentially important implications for gender inequality reproduction across generations. Causal interpretation is limited by cross-sectional data and potential fertility selection.

JEL Classification: J13, J16, I24, O15 **Keywords:** son preference, childcare, gender bias, parental investment, India

Data Availability: Analysis uses India Time Use Survey 2024 from National Statistical Office. Code available upon request.

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1 Introduction

India’s son preference is notorious: skewed sex ratios from sex-selective abortion, lower vaccination rates for girls, unequal food allocation favoring boys (Das Gupta et al. 2003; Sen 1990). Yet most research focuses on survival and health outcomes. We know less about whether son preference extends to **parental time investment**.

Does having sons lead parents to invest more time in childcare? If so, this represents a subtle but potentially consequential channel through which gender inequality reproduces across generations.

Main Hypothesis: Families with sons receive more total parental time than families with daughters, after controlling for family size and composition.

Data Innovation: We construct household child gender composition from individual-level records using relationship-to-head codes, linking to parental time use. This enables us to compare childcare time across households with only sons, only daughters, or mixed children.

Preview of Findings: We find that household child composition is associated with parental time allocation patterns, though effect sizes vary by parent gender and socioeconomic status. Cross-sectional analysis limits causal claims, as fertility decisions and child composition are endogenous.

1.1 Literature Review

Son preference in South Asia has been extensively documented across multiple domains. Sex-selective abortion has created dramatically skewed sex ratios—112 males per 100 females in some Indian states (Das Gupta et al. 2003). Girls receive lower vaccination rates, less nutritious food, and reduced medical care (Jayachandran and Kuziemko 2011).

Barcellos, Carvalho, and Lleras-Muney (2014) find that Indian parents invest more in boys’ education, with larger effects among wealthier families. This suggests son preference in

quality of investment, not just survival. Yet time use—the most immediate form of parental investment—remains understudied.

Parental time matters for child development (Cunha and Heckman 2008). Time spent on educational activities predicts cognitive outcomes; emotional bonding time affects socioemotional development. If parents systematically allocate less time to daughters, this creates early-life disadvantages that compound over time.

Our contribution: First analysis of son preference in parental time investment using nationally representative time use data with household child composition.

2 Data and Methods

2.1 Time Use Survey 2024

India’s Time Use Survey 2024, conducted by the National Statistical Office, collected 24-hour time diaries from 10.2 million individuals across 139,489 households. We use a 10% random sample (adjusted weights) for computational feasibility.

2.2 Household Child Gender Composition

Key Innovation: We construct household child composition from individual records using:

- **Relationship_to_Head** variable: Code 5 (unmarried child) or 6 (grandchild)
- **Gender:** 1=Male, 2=Female
- **Age:** < 18 years defines children

We aggregate to household level, counting:

- **num_sons:** Number of male children age < 18
- **num_daughters:** Number of female children age < 18
- **total_children:** Total children

We create three mutually exclusive categories:

- **Only sons:** 42.3% of households with children
- **Only daughters:** 34.0% of households
- **Mixed children:** 23.7% of households

2.3 Analysis Sample

We merge household composition to individual time use records and restrict to:

- **Married adults ages 25-50** (likely parents)
- **Households with children** (identified above)
- **Major activities only** (Major_Activity_Flag = 1)

Final sample: 180,997 person-day observations.

2.4 Key Variables

Dependent Variable:

- **childcare_minutes:** Minutes spent on childcare activities (Activity Code 31: caring for children, teaching children, physical care, accompanying children)

Independent Variables:

- **Household composition:** only_sons, only_daughters, mixed_children (reference)
- **Parent gender:** Female, Male
- **Controls:** Age, Age², Education, Urban, Employment status, State fixed effects, Total number of children

2.5 Estimation Strategy

We estimate:

$$childcare_i = \beta_0 + \beta_1 only_sons_h + \beta_2 only_daughters_h + \beta_3 female_i + \beta_4 female_i \times only_sons_h + X_i' \gamma + \delta_s + \epsilon_i$$

Where i indexes individuals, h indexes households, s indexes states. Standard errors clustered at household level.

Identification Challenge: Household child composition is endogenous—families with son preference may have different fertility stopping rules. We control for total children but cannot fully address selection. Results should be interpreted as associations, not causal effects.

3 Results

3.1 Descriptive Patterns: Childcare Time by Household Composition

Table 1: Parental Childcare Time by Household Child Gender Composition

Household Type	Minutes per Day		Participation Rate	
	Mother (min/day)	Father (min/day)	Mother (%)	Father (%)
NA	0.0	0.0	0.0	0.0
Mixed Children	12.2	0.2	24.4	0.3
Only Daughters	11.7	0.1	23.4	0.2
Only Sons	12.0	0.2	23.8	0.4

Pattern 1: Mothers' Time Allocation

Mothers in households with only sons spend 12.0 minutes per day on childcare, compared

to 11.7 minutes in households with only daughters. This represents a 2.9% difference (more time with sons).

Mothers in households with mixed children spend 12.2 minutes per day, falling between the two single-gender household types.

Pattern 2: Fathers' Time Allocation

Fathers in households with only sons spend 0.2 minutes per day on childcare, compared to 0.1 minutes with only daughters. This represents a 55.6% difference.

The gender gap in son preference is striking: fathers show 19.4 times larger proportional differences than mothers.

Interpretation: These descriptive patterns suggest household child composition is associated with parental time allocation. However, these are not causal estimates—unobserved factors (parental preferences, fertility histories) may confound the relationship.

3.2 Class Gradient in Son Preference

Table 2: Mother's Childcare Time by SES and Household
Child Composition

SES Group	Only Sons (min)	Only Daughters (min)	Gap (min)	Gap (%)
High SES	10.5	10.2	-0.2	-2.3
Low SES	11.8	12.2	0.4	3.4

Finding: Son preference in time investment varies by socioeconomic status.

Among high SES families (educated urban), the gap is 2.3%. Among low SES families (non-educated rural), the gap is 3.4%.

Interpretation: This socioeconomic gradient may reflect:

1. **Resources:** Wealthier families have more discretionary time to allocate differentially
2. **Aspirations:** Stronger expectations for sons' educational/career success among educated families
3. **Compensatory Effects:** Poor families may rely on daughters' domestic labor, reducing time gaps

3.3 Regression Analysis

Table 3: Childcare Time by Household Child Composition: Regression Analysis

	Dependent Variable: Childcare Minutes per Day			
	(1)	(2)	(3)	(4)
(Intercept)	0.333*** (0.091)	0.172*** (0.028)	-8.330*** (1.701)	
Only Sons	-0.126 (0.132)	0.022 (0.043)	-0.099 (0.083)	-0.016 (0.085)
Only Daughters	-0.351** (0.137)	-0.047 (0.035)	-0.128* (0.075)	-0.020 (0.077)
Female	11.775*** (0.091)	12.050*** (0.178)	10.325*** (0.214)	10.550*** (0.216)
Only Sons × Female		-0.253 (0.229)	-0.289 (0.230)	-0.228 (0.230)
Only Daughters × Female		-0.518** (0.235)	-0.536** (0.235)	-0.491** (0.235)
Age			0.626*** (0.090)	0.633*** (0.090)
Age ²			-0.008*** (0.001)	-0.008*** (0.001)
Higher Education			-0.584*** (0.146)	-0.406*** (0.144)
Urban			-0.255** (0.124)	0.006 (0.122)
Employed			-2.501*** (0.190)	-2.106*** (0.191)
Number of Children			-0.079*** (0.025)	-0.046* (0.025)
Num.Obs.	180 997	180 997	180 997	180 997
R2	0.082	0.082	0.084	0.086

Notes: Standard errors clustered at household level. Reference category: Households with mixed children (both sons and daughters). Model 4 includes state fixed effects.
* p < 0.1, ** p < 0.05, *** p < 0.01

Regression Results Interpretation:

Model 1 shows baseline associations without controls. Model 2 adds gender interactions. Model 3 adds individual-level controls. Model 4 adds state fixed effects.

Key Findings:

- Coefficients on **only_sons** and **only_daughters** capture effects for fathers (male reference group)

- Interaction terms capture *additional* effects for mothers
- The reference category is households with mixed children

Causal Interpretation: These are associations, not causal effects. Household child composition is endogenous—parents with strong son preference may have different fertility behaviors that also affect time use.

4 Discussion

4.1 Mechanisms: Why Son Preference in Time?

1. Educational Investment Hypothesis: Parents may view sons as better returns on educational investment, allocating time to homework help, tutoring, and enrichment activities.

2. Social Activities Hypothesis: Parents may invest time taking sons to sports, music lessons, and extra-curriculars while expecting daughters to help with housework.

3. Emotional Bonding Hypothesis: Cultural preference for sons may translate to more leisure and bonding time with male children.

4. Elder Care Expectations: Parents view sons as future economic support (daughters marry out), creating incentives for differential investment.

4.2 Intergenerational Implications

If son preference shapes early childhood time investment:

Human Capital Effects: Differential parental time can contribute to educational gender gaps, even when school enrollment is equal.

Socialization Effects: Daughters receiving less parental attention (while assigned more housework) internalize messages about their relative worth.

Inequality Reproduction: Early-life advantages for boys compound over time, perpetuating gender inequality across generations.

4.3 Policy Implications

1. **Make Time Investment Visible:** Public awareness campaigns highlighting how time allocation affects daughters' development.
2. **Father Involvement Programs:** Since fathers show larger gaps, interventions targeting paternal involvement with daughters could reduce disparities.
3. **Conditional Transfers:** Programs like Beti Bachao Beti Padhao could include conditions on parental time investment, not just school enrollment.

4.4 Limitations

1. **Cross-Sectional Data:** Cannot establish causality. Household composition is endogenous to fertility decisions.
2. **Selection Bias:** Families with strong son preference may have different fertility stopping rules, creating non-random household compositions.
3. **Single-Day Observation:** Time use captured for one 24-hour period. Multi-day diaries would better capture habitual patterns.
4. **Activity Aggregation:** We analyze total childcare time. Disaggregating by activity type (educational vs. routine care) would illuminate mechanisms.
5. **Child Age:** Cannot distinguish time allocation by child age. May be concentrated in specific developmental stages.
6. **Birth Order:** Cannot control for birth order effects, which may confound composition effects.

7. Household Clustering: While we cluster standard errors at household level, multiple individuals per household create dependencies.

5 Conclusion

Using India’s 2024 Time Use Survey linked to household child gender composition, we find suggestive evidence that household child composition is associated with parental time allocation patterns. The magnitude and direction of associations vary by parent gender and socioeconomic status.

These cross-sectional patterns are consistent with son preference extending from survival decisions to time investment. However, causal interpretation is limited by endogeneity of fertility and household composition. Longitudinal data following families over time would strengthen causal inference.

If confirmed by future research, these patterns have important implications for understanding how gender inequality reproduces across generations. Son preference may operate not just through who survives, but through how children are raised—creating early-life advantages for boys that compound into adult gender inequality.

6 References

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